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APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
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Title of Invention

CONTINUOUS TOURNAMENT-BASED COMPETITIVE SELECTION SYSTEM FOR AI STRATEGY OPTIMIZATION WITH AUTOMATED CHAMPION PROMOTION

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	4395	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74697802	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74697802.pdf	5	Application Data Sheet	109 KB
HS-2026-008-Final-Assigned-Updated-APP.TEXT.docx	4	Application body structured text document	38 KB

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
generatedADS74697802.pdf	77E2AE5F692B3CC9FE8362395775EE56FDF0A443D5CBC7E9

36F91D6CE56300C156DD2CAA74EE5BF39D0A6B3FDB5672A85BBF5C545317E9B9F9622D0FA47F76BA

HS-2026-008-Final-Assigned-Updated-APP.TEXT.docx

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

CONTINUOUS TOURNAMENT-BASED COMPETITIVE SELECTION SYSTEM FOR AI STRATEGY OPTIMIZATION WITH AUTOMATED CHAMPION PROMOTION

Title of the Invention	Continuous Tournament-Based Competitive Selection System for AI Strategy Optimization with Automated Champion Promotion
Docket Number	HS-2026-008
Inventor(s)	Eric Haywood
Residence(s)	149 Remington Street Unit 425, Fort Collins, CO 80524
Correspondence Address	HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524
Attorney/Agent (Name & Reg. No.)	None
U.S. Government Interest (Yes/No)	No
Gov. Agency & Contract No.	N/A

Title of the Invention

[0001] Continuous Tournament-Based Competitive Selection System for AI Strategy Optimization with Automated Champion Promotion

Cross-Reference to Related Applications

[0002] This application claims the benefit of priority and is related to co-pending provisional applications filed concurrently herewith by the same inventor and assignee.

Field of the Invention

[0003] This invention relates to systems and methods for continuously evaluating, comparing and selecting optimal AI processing strategies through automated tournament-style competitive elimination, with performance-based champion promotion to production environments.

Background of the Invention

[0004] Current AI systems select processing strategies using static configurations or simple A/B tests. These approaches fail to simultaneously evaluate more than two strategies, apply competitive pressure that drives continuous improvement, automatically promote winning strategies to production without human intervention, or detect champion performance degradation requiring re-evaluation. Multi-armed bandit approaches provide statistical optimization but lack structured elimination and promotion mechanisms needed for enterprise AI systems.

Summary of the Invention

[0005] The invention provides a system and method as described herein.

[0006] Tournament Engine: Runs continuous elimination tournaments where eight or more AI strategies compete head-to-head across multiple evaluation rounds, eliminating losers until a champion emerges.

[0007] Multi-Dimensional Scoring: Evaluates strategies across dimensions including accuracy, latency, throughput, cost efficiency and reliability, using Monte Carlo simulation to provide statistical confidence.

[0008] Elo-Style Rating System: Maintains persistent Elo ratings for each strategy across tournaments, enabling long-term performance tracking and seeding for future tournaments.

[0009] Automated Champion Promotion: Automatically promotes a strategy to production champion status when it wins a tournament and meets configurable performance thresholds, replacing the previous champion without human intervention.

[0010] Champion Degradation Detection: Continuously monitors the active champion's production performance and triggers re-evaluation tournaments when champion metrics fall below minimum thresholds.

Detailed Description of the Preferred Embodiment

A. Tournament Engine

[0011] The Arena Mode Service operates in continuous mode, running tournament cycles at a configurable frequency. Each tournament draws N participants (default eight) from a pool of registered AI strategies. Participants compete head-to-head in elimination rounds: for each round, Monte Carlo simulations evaluate each strategy across multiple dimensions, aggregate results into a composite score and eliminate the bottom performers until one champion remains.

[0012] Tournament-style Elo ranking generalizes pairwise comparisons to multi-agent tournaments and provides dynamic skill estimation. It iteratively updates each strategy's Elo rating based on head-to-head performance, enabling robust ranking across tournaments.

B. Elo Rating and Champion Promotion

[0013] After each tournament, Elo ratings for all participants are updated according to their performance relative to expectations. Strategies with higher ratings are seeded favourably in future tournaments.

[0014] When a strategy wins a tournament and exceeds a promotion threshold (for example, a win rate above 75 %, statistical confidence above 0.85 and a composite score exceeding the current champion's by at least 5 %), it is automatically promoted to production champion, replacing the previous champion without human intervention.

C. Degradation Monitoring

[0015] The system continuously monitors the active champion in production. If the champion's win rate drops below a demotion threshold or its performance metrics decline, the system triggers a new tournament to re-evaluate strategies. A challenger whose Elo rating surpasses the champion by more than 100 points may also trigger re-evaluation.

Claims

1. A computer-implemented method for AI strategy optimization, comprising: (a) maintaining a pool of AI processing strategies with persistent performance ratings; (b) conducting continuous elimination tournaments where strategies compete through Monte Carlo simulation across multiple evaluation dimensions; (c) updating persistent performance ratings for each strategy based on tournament results using an Elo-style rating system; (d) automatically promoting tournament winners to production champion status when performance exceeds configurable thresholds; and (e) continuously monitoring champion performance and triggering re-evaluation tournaments upon detecting degradation.

2. The method of claim 1, wherein the multiple evaluation dimensions comprise at least accuracy, latency, throughput, cost efficiency and reliability.

3. The method of claim 1, wherein each tournament round eliminates the bottom performing strategies based on Monte Carlo simulation results, continuing rounds until a single champion remains.

4. A system for continuous AI strategy selection comprising: a strategy pool maintaining registered processing strategies with Elo ratings; a tournament engine conducting elimination-style competitions using Monte Carlo simulation; a promotion engine that automatically deploys winning strategies to production; and a degradation monitor that detects champion performance decline and triggers re-evaluation.

Implementation Reference

[0016] The Arena Mode Service may be implemented in the file `src/services/arena-mode-service.js` with support from the `monte-carlo-service.js`. The service maintains Elo ratings and orchestrates elimination tournaments using Monte Carlo simulation across five or more performance dimensions.

References

[1] Tournament-style Elo ranking generalizes pairwise comparisons to multi-agent tournaments and provides dynamic skill estimation for AI models【930853352519900†L55-L70】

[2] Tournament-style ranking systems extend two-player Elo protocols to multi-agent competitions, allowing robust ranking across tournaments【930853352519900†L64-L74】

Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

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Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**CONTINUOUS TOURNAMENT-BASED COMPETITIVE SELECTION SYSTEM FOR AI STRATEGY OPTIMIZATION WITH AUTOMATED CHAMPION PROMOTION**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

HeadySystems Inc.
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Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	03/03/2026